REVERSE ENGINEERING – ACTIVITY REPORT

Product: Taksun TS-613A-2 Digital Clock (Car / Desk Clock with Alarm & Stopwatch)

Name(s): _____ Group: _____ Date: 06-10-2026

PART I – STUDYING THE PRODUCT

1. External Features

Physical Construct

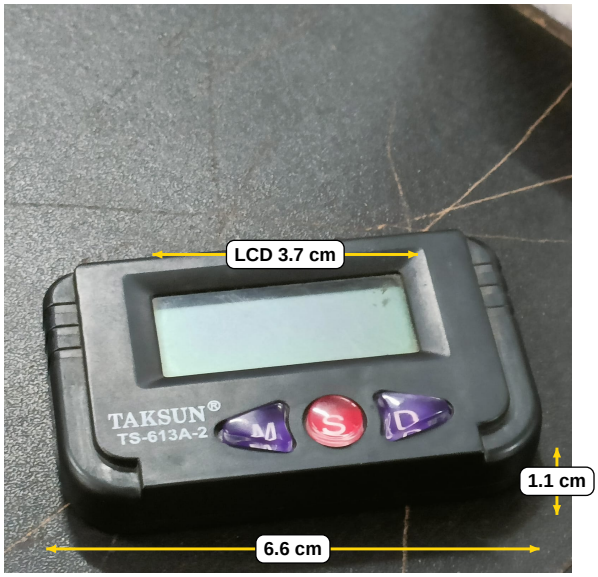


Fig 1. Front view (LCD + M, S, D keys)

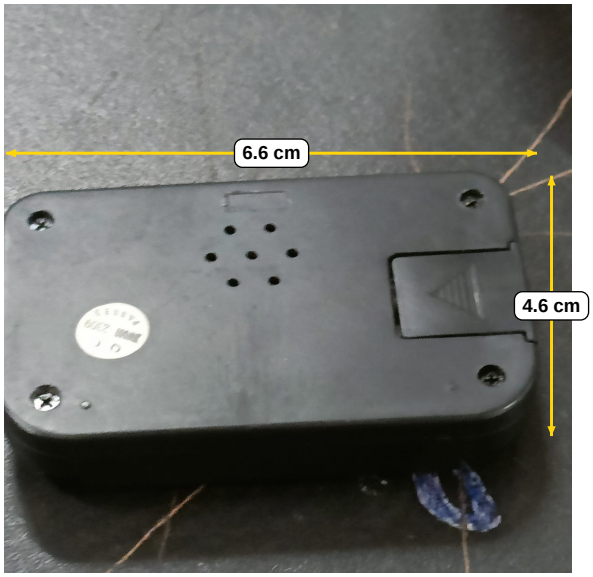


Fig 2. Back view (speaker holes, battery door, 4 screws)

• Compact, rectangular black ABS plastic enclosure (approx. 6-7 cm x 4 cm, pocket-size), made of a front bezel and a rear cover held by **4 Phillips screws**.

• Front: rectangular LCD window, branding “TAKSUN® TS-613A-2”, and three soft-touch keys: two purple triangular keys (**M** and **D**) and one round red key (**S**).

• Back: 7-hole speaker/buzzer grille, sliding battery-compartment door (arrow marks the opening direction), and a round “QC PASSED” sticker. A slot for a stand/adhesive mounting makes it suitable for dashboards and desks.

Subsystems Used

• Power supply (battery + contacts)

• Time-base (quartz crystal oscillator)

• Control / processing unit (clock IC, chip-on-board)

• Input unit (3 push keys with rubber keypad and PCB contact pads)

• Display unit (LCD with elastomeric “zebra” connector)

• Audio output (piezo buzzer)

• Mechanical enclosure

Physical dimensions (measured; marked on Fig 1 and Fig 2)

Length	Width	Thickness	LCD window width	Bezel steps
6.6 cm	4.6 cm	1.1 cm	3.7 cm	0.7 cm / 1.5 cm

Description of Subsystems

Subsystem	Description
Power supply	A small button cell sits in the back compartment; two metal spring contacts connect it to the PCB through red wires.
Time-base	A tiny cylindrical quartz crystal (32.768 kHz) on the PCB gives a stable reference frequency for timekeeping.
Control IC	A black epoxy blob on the PCB covers the bare silicon die (COB). It divides the crystal frequency, keeps time/date/alarm/stopwatch and drives the LCD.
Input unit	Rubber keypad with 3 nubs presses on carbon/gold contact pads on the PCB labelled M, S, D; plastic caps are the key tops.
Display	Monochrome 7-segment style LCD, mounted over a pink/grey conductive rubber strip that links its glass electrodes to the PCB pads.
Audio	Round piezo disc buzzer in the rear cover, connected by two red wires; sound exits through the grille holes.
Enclosure	Two-piece ABS shell with screw posts, key windows and LCD window.

Modes of Operation

• **Time mode** – shows hours:minutes (and seconds) and date/day.

• **Alarm mode** – set an alarm time; the buzzer sounds when time matches.

• **Stopwatch / timer mode** – start, stop and reset using the S key.

• **Setting mode** – M selects the mode / field, D and S change values (exact key function may vary slightly with the firmware; verify on the actual unit).

2. Purpose of the Product

The Taksun TS-613A-2 is a low-cost, battery-powered digital clock whose purpose is to display accurate time and date, sound an alarm and act as a stopwatch, in a small portable form that can be stuck on a car dashboard or kept on a desk.

3. User Needs Addressed

• Easy-to-read time and date at a glance (especially in vehicles that lack a clock).

• Portability and low power: runs for months on one small cell.

• Alarm / reminder and stopwatch functions in one device.

• Very low cost and simple three-key operation.

• Compact size and easy mounting.

PART II – PRODUCT ORIGIN TO LATEST INVENTIONS

1. Difference in Technology Across Versions

Key improvements and drawbacks at each stage of time-keeping technology that led to this product:

Stage	Technology	Key improvements	Drawbacks
1.1 Early stage	Sundials, water clocks, mechanical and pendulum clocks (Huygens pendulum clock, 1656)	First reliable time measurement; pendulum greatly improved accuracy	Bulky, needed winding, affected by temperature and vibration, drift of seconds per day
1.2 First generation	Quartz crystal clock (Warren Marrison, Bell Labs, 1927) and analog quartz watches (Seiko Astron, 1969)	Much higher accuracy (seconds per month), no winding	Early quartz clocks were large and expensive; analog display with motor and gears
1.3 Second generation	LED digital watches (Hamilton Pulsar, 1972) and early LCD watches (Seiko, 1973)	No moving parts, digital display, LCD very low power	LED drained battery fast and needed a button press; early LCDs had poor contrast/viewing angle; multi-chip circuits
1.4 Third generation (this product)	CMOS clock IC in chip-on-board form, 32.768 kHz crystal, LCD, piezo buzzer	Single-IC integration, very low cost and power, alarm + stopwatch, months of battery life	No backlight, no connectivity, accuracy depends on crystal (typically tens of seconds/month), manual setting
1.5 Latest generation	Smart clocks / smartwatches: RTC with temperature compensation, Bluetooth/Wi-Fi/NTP/GPS sync, radio-controlled clocks, OLED/e-ink displays, app control	Auto time sync and time-zone update, notifications, sensors, rechargeable battery	Higher cost, more power, needs charging, software dependency

2. Current Status – Core Smart Technologies Used in the Product

• **Quartz oscillation** (32.768 kHz) as the precise time-base, divided by 2^{15} to get 1 Hz.

• **CMOS ultra-low-power IC** in chip-on-board packaging for compact, low-cost integration.

• **LCD with multiplexed segment drive** using a low-power elastomeric connector.

• **Piezoelectric buzzer** for loud alarm at very low current.

• **Rubber-dome / membrane keypad** for sealed, cheap user input.

This product is not “smart” in the connected sense; it represents the mature, low-power embedded generation of clocks, and it has no wireless features.

3. Engineering Importance of This Evolution

• Accuracy improved from seconds/day (mechanical) to seconds/month (quartz).

• Power consumption fell from mechanical energy or mA (LED) to microamps (CMOS + LCD).

• Size, weight and cost dropped due to integration (many components into one IC).

• Reliability increased by removing moving parts.

• Added functions (alarm, stopwatch, calendar) became possible through digital logic, and today’s smart devices extend this with connectivity and sensors.

PART III – INTERNAL COMPONENTS & WORKING FLOW

1. Introduction of the Product

The Taksun TS-613A-2 is a battery-operated digital LCD clock with alarm and stopwatch. Internally it is a very simple embedded system: one crystal oscillator, one clock IC, an LCD, three keys and a buzzer, all on a single small PCB (about 38.5 mm x 27.5 mm, silkscreen “VER 2.0”, dated 18.04.20).

2. Photograph of the Internal Components of the Product

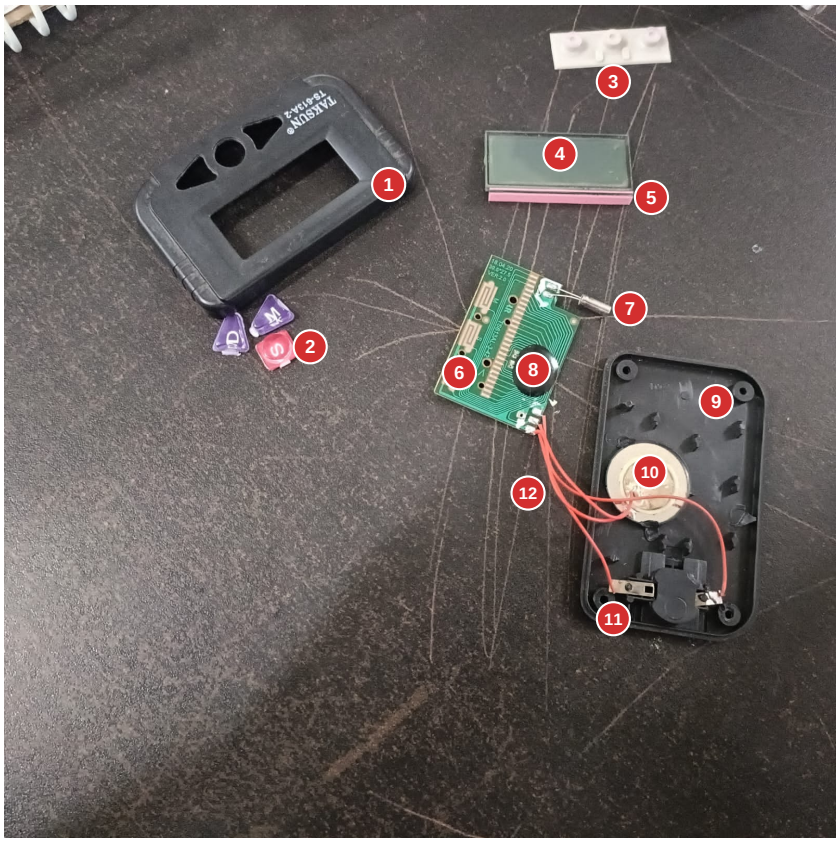


Fig 3. Disassembled clock: front bezel, key caps (D, M, S), rubber keypad strip, LCD with conductive strip, green PCB with crystal and epoxy IC, rear cover with piezo buzzer and battery contacts.

Component names (numbered on Fig 3)

No.	Component	Role in the product
1	Front bezel (ABS)	Holds LCD and keys; frames the display
2	Key caps (M, S, D)	Plastic key tops pressed by the user
3	Rubber keypad strip	Three nubs that close the PCB contact pads
4	LCD module	Monochrome 7-segment display of time/date
5	Conductive (zebra) strip	Links LCD glass electrodes to PCB pads
6	PCB (green, VER 2.0)	Single-layer board carrying IC, crystal, key pads
7	Quartz crystal 32.768 kHz	Accurate time-base for the clock IC
8	Clock IC (COB epoxy blob)	Counts time, drives LCD and buzzer
9	Rear cover (ABS)	Holds buzzer and battery contacts; 4 screws
10	Piezo buzzer disc	Produces alarm / key-beep sound
11	Battery contacts	Metal spring contacts for the button cell
12	Connecting wires (red)	Carry power and buzzer signal to the PCB

3. Internal Block Diagram (Functional View)

Power --> Timing --> Processing --> Output

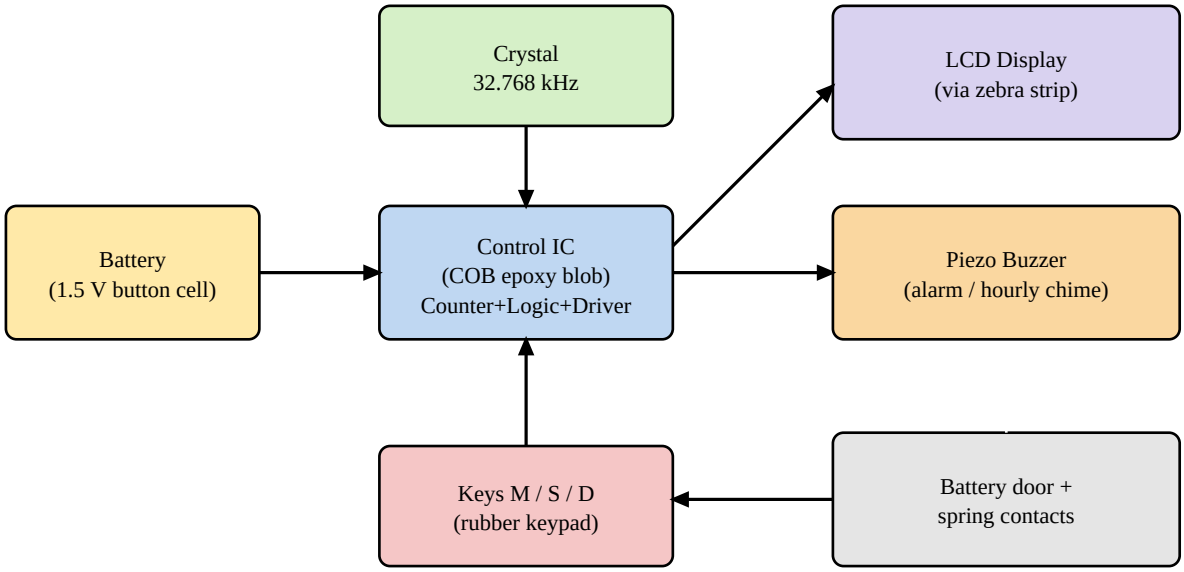


Fig 4. Functional block diagram

4. Internal Subsystems with Detailed Explanation

Battery & contacts (power)

Point	Explanation
4.1 Function (why used?)	Supplies the DC voltage to the IC, display and buzzer so the clock runs continuously.
4.2 Working principle	Chemical energy of the cell is delivered through metal spring contacts and wires to the PCB supply pads.
4.3 Characteristics	Nominal 1.5 V, low capacity, very small current (microamps in time mode, higher during alarm).
4.4 Role / need in this product	Needed for portability; the IC is CMOS and can run for months on this small cell.

Quartz crystal (time-base)

Point	Explanation
4.1 Function (why used?)	Generates a precise, stable frequency so time is kept accurately.
4.2 Working principle	Quartz is piezoelectric; with the oscillator circuit in the IC it vibrates at 32,768 Hz. The IC divides it by 2^{15} to get 1 Hz (one second).
4.3 Characteristics	Frequency 32.768 kHz, typical tolerance about +/-20 ppm, tuning-fork cylindrical package, low power.
4.4 Role / need in this product	Without a crystal a simple RC oscillator would drift minutes per day.

Clock IC (COB)

Point	Explanation
4.1 Function (why used?)	Brain of the clock: counts seconds, minutes, hours, date, alarm and stopwatch, scans keys and drives LCD and buzzer.
4.2 Working principle	Divider chain + counters + decoders generate segment data; the IC multiplexes the LCD (backplane/segment signals) and produces buzzer tones.
4.3 Characteristics	CMOS, low voltage (about 1.5 V), ultra-low power, bare die covered by epoxy (chip-on-board).
4.4 Role / need in this product	Integrates what once needed many chips, reducing cost and size.

LCD with zebra connector

Point	Explanation
4.1 Function (why used?)	Displays time, date and mode information.
4.2 Working principle	Liquid crystal twists light polarisation when an AC voltage is applied across a segment; a conductive rubber strip links the glass electrodes to the PCB pads.
4.3 Characteristics	Monochrome, reflective, 7-segment style, very low power (microwatts), visible in daylight, no backlight.
4.4 Role / need in this product	Cheapest low-power readable display for a clock.

Keys and keypad

Point	Explanation
4.1 Function (why used?)	Allow the user to set time, change mode and start/stop the stopwatch.
4.2 Working principle	Pressing a rubber nub shorts the PCB contact pads; the IC detects the change and acts on it.
4.3 Characteristics	Three momentary keys (M, S, D), contact resistance a few hundred ohms, thousands of presses life.

Point	Explanation
4.4 Role / need in this product	Simple, sealed and low-cost input for a few functions.

Piezo buzzer

Point	Explanation
4.1 Function (why used?)	Produces alarm / key beep sound.
4.2 Working principle	A piezo ceramic disc on a metal plate bends when voltage is applied; an oscillating signal from the IC makes it vibrate and produce sound.
4.3 Characteristics	Typically 2-4 kHz resonance, low current, loud for its size.
4.4 Role / need in this product	Needs no coil/magnet, so it is cheap and thin.

5. Complete Working Flow (Step-by-Step)

1. Battery is inserted; supply reaches the clock IC and the oscillator starts.

2. The 32.768 kHz crystal oscillates and the IC divides this frequency down to 1 Hz.

3. Every second the counters update seconds, minutes, hours, day and date.

4. The IC converts the count to 7-segment data and drives the LCD segments, so the time is displayed.

5. When a key (M/S/D) is pressed, the rubber nub closes the PCB contacts and the IC changes mode or edits values.

6. In alarm mode the IC compares the current time with the alarm time; on match it sends a tone signal to the piezo buzzer.

7. In stopwatch mode the IC counts elapsed time and shows it until it is stopped or reset.

6. Components List – Bill of Material

S.No	Component	Qty
1	Front bezel (ABS plastic)	1
2	Rear cover (ABS plastic) with 4 screws	1 + 4
3	Key caps (M, S, D)	3
4	Rubber keypad strip (3 nubs)	1
5	LCD module	1
6	Conductive rubber (zebra) strip	1
7	PCB (green, VER 2.0)	1
8	Clock IC (COB, epoxy blob)	1
9	Quartz crystal 32.768 kHz	1
10	Piezo buzzer disc	1
11	Battery contacts (metal springs/plates) and connecting wires	2 + 2-3 wires
12	Button cell battery	1
13	Battery door and QC sticker	1 + 1

7. Component Ratings & Specifications

Component	Typical rating / specification
Battery	1.5 V button cell (e.g. AG13/LR44 type); check the cell fitted
Crystal	32.768 kHz, tuning-fork type, about +/-20 ppm
Clock IC	CMOS, about 1.2-1.5 V supply, microamp standby current
LCD	Monochrome TN type, 7-segment, static or multiplexed drive
Piezo buzzer	About 3 V peak drive signal, resonance around 2-4 kHz
PCB	About 38.5 mm x 27.5 mm, single layer, marked VER 2.0 (18.04.20)
Keys	Momentary rubber contact type, 3 keys

Note: ratings are typical values for this class of component (not printed on the parts). Confirm with a multimeter / datasheet where required.

8. Component Function Table

Component Name	Function	Specifications
Battery	Power source	1.5 V button cell
Battery contacts	Connect cell to PCB	Spring metal contacts
Crystal	Accurate time reference	32.768 kHz
Clock IC (COB)	Counting, control, display and buzzer drive	Low-power CMOS
LCD	Shows time/date/mode	7-segment, monochrome
Zebra strip	Links LCD to PCB	Conductive rubber
Keypad + caps	User input (M, S, D)	3 momentary keys
Piezo buzzer	Alarm sound	Piezo disc, 2-4 kHz
Enclosure	Protection and mounting	ABS, 4 screws

PART IV – TINKERCAD SIMULATION OF THE DIGITAL CLOCK

1. Aim of the Simulation

To model the working of the Taksun TS-613A-2 in software before building it: an Arduino Uno plays the role of the clock IC (counting and control), a 16x2 LCD plays the role of the display, and three push buttons replace the M, S and D keys. The circuit was built and tested in Tinkercad Circuits.

2. Circuit Diagram and Output

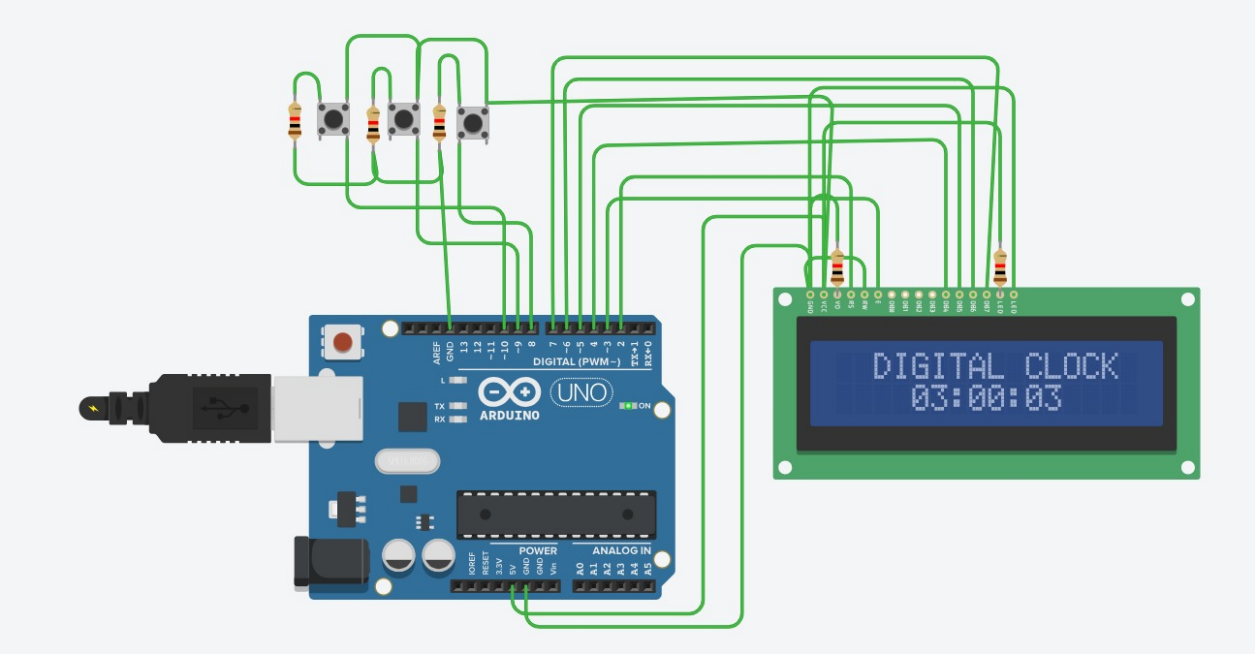


Fig 5. Tinkercad circuit: Arduino Uno, 16x2 LCD (showing "DIGITAL CLOCK 03:00:03") and three push buttons with resistors

3. Components Used and Connections (as per the code)

Component	Qty	Connection / purpose
Arduino Uno	1	Microcontroller; counts time with millis() and drives the display (replaces the clock IC)
16x2 LCD (HD44780 type)	1	RS = pin 2, E = pin 3, D4–D7 = pins 4–7 (4-bit mode); shows title and HH:MM:SS (replaces the LCD)
Push buttons	3	Pin 10 = increment hours, pin 11 = increment minutes, pin 12 = reset seconds (replace the M / D / S keys)
Resistors	3 + 2	Pull-down resistors on the three buttons; two resistors on the LCD contrast (VO) and backlight (LED) pins
Power / ground	–	5 V and GND from the Arduino to the LCD and button network

4. Working of the Simulation

- The clock starts at 12:00:00. Every 1000 ms the program adds one second and rolls over seconds to minutes and minutes to hours (12-hour format).
- The first LCD row shows the fixed title "DIGITAL CLOCK"; the second row shows the time and is refreshed after every change.
- Pressing a button reads HIGH on its pin: hours or minutes are incremented, or seconds are reset to 00. A 250 ms delay prevents the value from skipping.

PART V – ARDUINO CODE USED IN THE SIMULATION

Blank lines were removed only to fit the code on one page; the program logic is unchanged.

```
#include <LiquidCrystal.h>
// Initialize LCD with your specific circuit pin arrangement
LiquidCrystal lcd(2, 3, 4, 5, 6, 7);
// Button configurations
const int button1Pin = 10; // Press to increment HOURS
const int button2Pin = 11; // Press to increment MINUTES
const int button3Pin = 12; // Press to RESET seconds to 00
// Time tracking variables
int hours = 12;   // Clock starts at 12:00:00
int minutes = 0;
int seconds = 0;
// Variables to handle internal timing without stopping the processor
unsigned long previousMillis = 0;
const long interval = 1000; // 1000 milliseconds = 1 second

void setup() {
  lcd.begin(16, 2);
  pinMode(button1Pin, INPUT);
  pinMode(button2Pin, INPUT);
  pinMode(button3Pin, INPUT);
  // Static top label similar to a digital desk clock
  lcd.setCursor(0, 0);
  lcd.print("  DIGITAL CLOCK ");
}

void loop() {
  unsigned long currentMillis = millis();
  // --- AUTOMATIC TIME KEEPING ---
  // Checks if 1 full second has naturally passed
  if (currentMillis - previousMillis >= interval) {
    previousMillis = currentMillis;
    seconds++;
    if (seconds >= 60) {
      seconds = 0;
      minutes++;
    }
    if (minutes >= 60) {
      minutes = 0;
      hours++;
    }
    if (hours > 12) { // 12-hour format structure
      hours = 1;
    }
    updateClockDisplay();
  }
  // --- MANUAL BUTTON CONTROLS (Taksun Adjustment Mode) ---
  // Button 1: Adjust Hours manually
  if (digitalRead(button1Pin) == HIGH) {
    hours++;
    if (hours > 12) hours = 1;
    updateClockDisplay();
    delay(250); // Prevents rapid skipping
  }
  // Button 2: Adjust Minutes manually
  if (digitalRead(button2Pin) == HIGH) {
    minutes++;
    if (minutes >= 60) minutes = 0;
    updateClockDisplay();
    delay(250);
  }
  // Button 3: Reset Seconds to zero
  if (digitalRead(button3Pin) == HIGH) {
    seconds = 0;
    updateClockDisplay();
    delay(250);
  }
}

// Function to properly format numbers (adds leading zero if less than 10)
void printDigits(int digits) {
  if (digits < 10) {
    lcd.print('0');
  }
  lcd.print(digits);
}

// Refreshes the display interface layout
void updateClockDisplay() {
  lcd.setCursor(0, 1);
  lcd.print("    "); // Indent center spacing
  printDigits(hours);
  lcd.print(":");
  printDigits(minutes);
  lcd.print(":");
  printDigits(seconds);
  lcd.print("    ");
}
```

CONCLUSION

This activity studied the Taksun TS-613A-2 digital clock from its outside to its inside. Externally it is a compact black ABS enclosure of 6.6 cm x 4.6 cm x 1.1 cm with an LCD window 3.7 cm wide and three keys (M, S, D). On opening it, the product was found to be a very simple embedded system: a 1.5 V button cell, a 32.768 kHz quartz crystal, a chip-on-board clock IC, an LCD connected through a conductive zebra strip, a rubber keypad and a piezo buzzer, all on one small green PCB (VER 2.0).

The evolution study showed how time-keeping moved from mechanical and pendulum clocks to quartz, then to LED/LCD digital watches and finally to low-cost CMOS clock ICs and smart, connected devices. This progress improved accuracy from seconds per day to seconds per month, reduced power use to microamps, and cut size and cost through integration. The TS-613A-2 represents the mature, low-power generation of this evolution; it has no backlight or wireless time sync.

To understand the working flow, the clock was recreated in Tinkercad with an Arduino Uno, a 16x2 LCD and three push buttons. The simulation reproduces the main function of the product: it keeps time every second, shows it on the display and lets the user adjust hours, minutes and seconds with buttons. Comparing the two shows how the original product integrates in one IC what the prototype does with a microcontroller and separate parts.

Comparison: actual product vs. simulation

Function	Taksun TS-613A-2	Tinkercad simulation
Time-base	32.768 kHz quartz crystal, divided to 1 Hz	Arduino clock with millis() (1000 ms interval)
Control	COB CMOS clock IC	ATmega328P on Arduino Uno
Display	Monochrome 7-segment LCD via zebra strip	16x2 character LCD (4-bit mode)
Input	3 rubber keys: M, S, D	3 push buttons (hours, minutes, reset seconds)
Alarm / stopwatch	Piezo buzzer, alarm and stopwatch modes	Not implemented (time display and setting only)
Power	1.5 V button cell, months of life	5 V from USB / Arduino

Learning and Future Scope

- Reverse engineering showed how each subsystem (power, time-base, control, input, display, audio) contributes to the product and why the design is low-cost and low-power.
- The simulation can be extended with an alarm and a buzzer, a stopwatch mode, an AM/PM indicator and a real-time clock module (e.g. DS3231) for better accuracy.
- The product can be improved with a backlight, automatic time sync and a rechargeable cell, at the cost of higher price and power use.